Custom Reinforcement Learning Agent Implementation Using Farama Gymnasium

Technical report template • Taxi-v3 • Tabular Q-learning • [INSERT STUDENT NAME / REGISTRATION NUMBER]

1. Problem Description

The Taxi-v3 environment represents a passenger delivery problem on a 5 × 5 grid. Its 500 encoded states combine five taxi rows, five columns, five passenger positions/status values and four possible destinations (5 × 5 × 5 × 4). The six actions are south (0), north (1), east (2), west (3), pickup (4) and drop-off (5). The standard rewards are −1 per regular step, −10 for an illegal pickup/drop-off, and +20 for a successful delivery. Delivery terminates an episode; otherwise the standard time limit truncates it after 200 steps. The goal is to learn a policy maximizing discounted cumulative reward while avoiding unnecessary actions.

2. Methodology

A Q-table of size 500 × 6 is initialized with zeros. One table entry is updated after every transition using off-policy temporal-difference learning:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

Here s is the current state, a the executed action, r the immediate reward, s′ the next state, a′ a possible next action, Q(s,a) the estimated action value, α the learning rate and γ the discount factor. In a genuinely terminated state, the future-value term equals zero; time-limit truncation ends the trajectory but does not make the next state terminal for the Q-learning target.

Epsilon-greedy exploration chooses an action at random with probability ε and otherwise chooses an action maximizing Q. After each training episode, $\varepsilon_{k+1} = \max(\varepsilon_min, \varepsilon_k \times d)$, where d controls multiplicative decay. Q-value ties are randomly resolved during training; evaluation uses a deterministic greedy policy with no exploration or updates.

The implementation is modular: agent.py implements Q-values, action selection and updates; utils.py provides moving averages, plots, greedy evaluation and GIF export; main.py manages environment interaction, seed control and saved outputs; and experiments.py performs controlled hyperparameter sensitivity runs. The notebook documents the entire workflow.

3. Experimental Setup and Hyperparameters

The values below are an initial baseline, not experimentally tuned findings. Enter selected final settings and keep training/evaluation budgets identical when comparing configurations.

Hyperparameter	Initial baseline	Selected final setting
Environment	Taxi-v3	Taxi-v3
Training episodes	15,000	[INSERT FINAL COUNT]
Learning rate α	0.10	[INSERT SELECTED α]
Discount factor γ	0.95	[INSERT SELECTED γ]
Initial ε	1.00	[INSERT START ε]
Minimum ε	0.01	[INSERT MINIMUM ε]
Epsilon decay d	0.9995	[INSERT SELECTED d]
Main seed / evaluation	42 / 100 episodes	[INSERT VALUES]

A one-factor-at-a-time study compares $\alpha \in \{0.05, 0.10, 0.50\}$, $\gamma \in \{0.90, 0.95, 0.99\}$, and $d \in \{0.999, 0.9995, 0.9998\}$, each with seeds 0, 1 and 2. Other hyperparameters remain fixed during each sweep. Per-seed greedy evaluation metrics and mean smoothed training curves ± one standard deviation are recorded. No performance numbers are populated in this template.

4. Results and Discussion

FIGURE 1 PLACEHOLDER: Insert outputs/training_curves.png here. Rewards, steps and epsilon versus episode.

FIGURE 2 PLACEHOLDER: Insert alpha, gamma and epsilon_decay comparison charts. Use legible multi-panel figures or reference an appendix within 4 pages.

For convergence, define a criterion based on sustained reward/episode-length stability, not a single favorable episode. Estimated convergence: [INSERT EPISODE OR NOT CLEARLY OBSERVED].

4. Results and Discussion (continued)

Greedy-policy evaluation metric	Observed result
Mean reward / episode	[INSERT FINAL MEAN REWARD FROM MY RUN]
Successful deliveries (%)	[INSERT SUCCESS RATE FROM MY RUN]
Average steps / episode	[INSERT AVERAGE STEPS FROM MY RUN]
Convergence episode, if identifiable	[INSERT EPISODE OR UNCLEAR]
Chosen α, γ and d	[INSERT VALUES + EVIDENCE]

Describe the measured direction and magnitude of changes in reward and steps across training: [INSERT OBSERVATION FROM FIGURE 1]. Explain which learning-rate, discount-factor and decay choices improved held-out greedy evaluation and whether the ± standard-deviation bands indicate stable behavior across seeds: [INSERT EVIDENCE FROM FIGURE 2]. Be precise if curves overlap or a configuration does not clearly outperform the baseline.

Potential failure modes include illegal pickup/drop-off operations, repeated movements, unvisited state-action pairs and reaching the 200-step limit. Excessively large α may produce volatile value updates; a very small γ may underweight distant delivery reward; overly rapid ε decay may prevent necessary exploration. These are possible explanations, not claimed experimental observations. If any occurred, identify the corresponding curve, recorded actions, or failure episodes. The demonstration GIF is a successful example trajectory, not evidence of overall success probability.

5. Conclusion

This study implements tabular Q-learning for the 500-state Taxi-v3 environment using a handwritten value update and epsilon-greedy behavior rather than a prebuilt RL library. Learning progress is assessed with reward, step-length and exploration curves, while policy performance is assessed by separate greedy evaluation over multiple episodes. Sensitivity to α, γ and ε decay is assessed using repeated, one-factor-at-a-time experiments. The evidence supports the following conclusion: [INSERT SHORT CONCLUSION GROUNDED IN YOUR MEASURED RESULTS]. Future improvements may explore alternative exploration schedules, action masking, or additional random seeds.

References

Farama Foundation. Gymnasium Taxi environment documentation, https://gymnasium.farama.org/v1.2.3/environments/toy_text/taxi/

Sutton, R. S., and Barto, A. G. Reinforcement Learning: An Introduction (2nd ed.). MIT Press, 2018.

Completion note: Replace all bracketed fields and figure boxes with your own executed results. Keep the completed report within four pages.